DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II Advanced Calculus (2AC)

Tutorial Sheet 7: **6th April 2023**

1. Find the cylindrical coordinates (r, θ, z) and the spherical coordinates (ρ, θ, ϕ) of the point with cartesian coordinates $\left(\frac{\sqrt{2}}{8}, \frac{\sqrt{6}}{8}, \frac{\sqrt{2}}{4}\right)$. Explain clearly how you found those coordinates.

2. Describe the given sets in the system of cartesian coordinates:

   (a) $A : \phi = \frac{\pi}{3}$ in spherical coordinates; (b) $B : \frac{\pi}{2} \leq \theta \leq \pi$ in cylindrical coordinates;
   (c) $C : \theta = \frac{\pi}{4}$ in cylindrical coordinates; (d) $D : \theta = \frac{\pi}{4}$ in spherical coordinates.

3. A torus $\mathbb{T}$ is obtained by rotating a circle C in the xz-plane about the z-axis. If C has radius r and center $(R, 0, 0)$ with $R > r$ then the parametrization of the torus is given by:

$$\boldsymbol{r}(s,t) = ((R + r\cos s)\cos t, (R + r\cos s)\sin t, r\sin s) \quad 0 \leq s \leq 2\pi \text{ and } 0 \leq t \leq 2\pi\,.$$

   (i) Show that the torus $\mathbb{T}$ is a regular surface.

   (iii) Write down the cartesian equation of the tangent plane to the torus $\mathbb{T}$ at the point $(0, R, r)$.

4. Convert in cartesian and cylindrical coordinates the equation of the surface $\mathcal{S}$ given in sperical coordinates by $\rho = 3$.

5. Let $\mathcal{S}$ be the surface described by the vector function $\vec{r} : [0, \frac{3\pi}{2}] \times [\frac{\pi}{4}, \frac{2\pi}{3}]$ defined by

$$\vec{r}(s,t) := (2\sin s \sin t, 3\cos t, \cos s \sin t) \quad \forall (s,t) \in \left[0, \frac{3\pi}{2}\right] \times \left[\frac{\pi}{4}, \frac{2\pi}{3}\right].$$

   (a) Show that the surface $\mathcal{S}$ is regular.

   (b) Write down a cartesian equation of the surface $\mathcal{S}$.

   (c) Write down the cartesian equations of the tangent plane to the surface $\mathcal{S}$ at the point $(0, 3/2, \sqrt{3}/2)$.

6. Given a differentiable function $f : \mathbb{R}^3 \to \mathbb{R}$. We recall that the normal to the tangent plane of any level surface $f(x,y,z) = k$ at the point (x_0, y_0, z_0) is given by $\nabla f(x_0, y_0, z_0)$.
   Find a cartesian equation for the tangent plane to the surface $x^2 + 2xy - y^2 + z^2 = 2$ at the point $(-1, 2, -3)$.